MICROPROSE · 1994 · MS-DOS

Sid Meier's COLONIZATION

Technical Reference

The shipped MS-DOS release, documented from its binaries and data files. Executables and overlay architecture · asset containers · terrain and the map compositor · colonies, market, and the native economy · units, movement, and combat · powers, diplomacy, and revolution · events and strings · the UI engine, screen by screen · the map editor · data structures.

RECONSTRUCTED BY DISASSEMBLY OF VICEROY.EXE AND MAPEDIT.EXE · EVERY FIGURE TRACED TO A FILE OFFSET, A DATA-FILE FIELD, OR LIVE GAME MEMORY — OR MARKED UNMAPPED

Contents

The shipped 1994 MS-DOS release, documented from its binaries and data files. Facts in this volume were established by disassembly of the shipped executables, by reads of live game memory under emulation, and by pixel-level comparison of screens reconstructed from these pages against the running game. Where a value could not be established it is marked unmapped or runtime — no figure in this volume is invented.

PART V

Politics and powers

§19 Natives

Eight tribes populate the map with individually tracked villages. This chapter covers the whole native system in one place: the variables first, then every interaction in play order — meeting a tribe, entering its villages, trading, anger and alarm, missions, demands and hired wars, raids, and finally attacking and burning settlements. Everything here traces to the shipped binary or the original manual; where the binary has not yet given up a number, the item says so plainly (TBD) instead of guessing.

19.1 The native state model — every variable

The engine keeps native state in three layers:

1. **One record per tribe** (eight live records) — the tribe's advancement level, its life/death status, its trade stocks, and the per-power anger seed rolled at map generation.
2. **One 18-byte record per village** (up to 84, live count in `game.settlement_count`) — position, owner, population, mission, trespass memory, trade memory, and a per-European-power **alarm** word.
3. **A per-(tribe, power) tension table** — a 0..100 anger meter read through `get_tension()` and written only by the single applier `adjust_tension()`.

While a native event is being processed the engine keeps four context variables current: the active settlement record, the active tribe record, the tribe's index, and the tribe's power id (always the tribe index + 4).

settlement — the record's variables storage layout in Appendix A - binding in Appendix C

VARIABLE	EXAMPLE	MEANING
`settlement.map_x`	23, 41	tile position
`settlement.map_y`	↑	
`settlement.owner`	7 = Iroquois	power id 4..11 = tribe index + 4
`settlement.flags`	4 = capital	`0x04` capital; `0x02` taught; `0x01` rebuild-a-brave; `0x10` tribute-once
`settlement.population`	8	village size — grows to the tribe cap; −1 per lost battle
`settlement.mission`	255 = no mission	`0xFF` none; low nibble owner; `0x10` expert (Brébeuf)
`settlement.growth_counter`	3	+= population each turn; acts at 20 (grow or spawn)
`settlement.trade_marker`	254 = trespass stamped	gift/trespass memory: the gifted good, or the trespass stamp
`settlement.last_sold`	255 = nothing yet	last good the player sold to the village
`settlement.last_bought`	2 = Tobacco	last good the player bought from it
`settlement.alarm[4]`	England 40 · France 12 · Spain 0 · Neth. 5	per-power anger words — war and raids at 128+

settlement.flags

7	6	5	4	3	2	1	0
0x80	0x40	0x20	0x10	0x08	0x04	0x02	0x01
					capital	taught	w/o

capital — set; doubles value

taught — already taught

w/o — write-only bit

settlement.mission 255 = no mission

7	6	5	4	3	2	1	0
0x80	0x40	0x20	0x10	0x08	0x04	0x02	0x01
			expert	owning power			

owning power — European power holding the mission
expert — expert-mission doubler (Brebeuf; set, tested)

tribe — the record's variables storage layout in Appendix A - binding in Appendix C

VARIABLE	EXAMPLE	MEANING
`tribe.capital_x`	23, 41	the capital's tile
`tribe.capital_y`	↑	
`tribe.level`	2 = Advanced (Aztecs)	tech level 0..3 — the camp/village/city class
`tribe.status`	128 = tribe wiped out	0x20 armed for war; 0x40 avenge-a-razing; 0x80 wiped out
`tribe.muskets_known`	5	musket lore — an armory that depletes arming braves
`tribe.horses_known`	3	horse lore — a pure breeding rate, never spent
`tribe.horses_stock`	40	the herd: +horses_known per turn, cap 2·(tribe pop+25)
`tribe.silver_stock`	120	map-creation endowment (no runtime reader — TBD)
`tribe.goods_stock[16]`	Furs 60 · Tobacco 25…	net trade balance per good; drifts to 0 by level+1 per turn
`tribe.requests[4]`	—	per-power request words, cleared each turn (codes TBD)
`tribe.tension_frac[4]`	England +5 of ±8	fractional anger accumulator — every ±8 is one tension tick
`tribe.tension[4]`	England 40 · France 12 · Spain 0 · Neth. 5	per-power anger 0..100 — THE tension table, seeded at map creation

The two anger meters. They are separate and both matter:

- `settlement.alarm[power]` — a per-village word. At **128 or more** the village is on a war footing toward that power: it becomes a raid source and poisons colony-site desirability. This is also the number the village-entry code uses to swap *Trade With Village* for *Enter Hostile Village* at **75**.
- **tension(tribe, power)** — the 0..100 tribe-level meter. **75 and above is hostile; 100 is war.** Every change goes through `adjust_tension()`, which clamps the result to 0..100 and **halves every positive (angering) delta** for France and for any power that has **Pocahontas** in Congress. (The applier's fourth "category" argument is passed by all 33 callers and read by none — it is dead.)

The visible **attitude phrase** (Content / Uneasy / Restless / Angry, dressed with Extremely / Very / Rather / Somewhat / Slightly) is a display banding of a colonial-presence score at cutoffs −5 / 0 / 10; *War* is the separate alarm ≥ 128 state. The presence score's exact composition is multi-term and not fully decomposed — TBD.

Data-model corrections carried by this chapter (2026-08-01 ruling): the tension table physically lives at the tail of each tribe record — it is per-tribe, not per-village; the village growth counter is real (an earlier "never accessed" verdict came from a scan that could not see pointer-relative access); the alarm block is four 16-bit words; and bytes 7–9 are the village's traded-good memory, not a wealth account.

19.2 The eight tribes

Tribe identity loads from NAMES.TXT `@TRIBES` at game start. The **color** column is the tribe's palette index — the tint of its orders boxes and map markings, exactly parallel to the European powers' colors (England 12, France 9, Spain 14, Netherlands 13) per the 2026-08-01 ruling. The **settlement** column shows the map art each tribe's class draws: camps for semi-nomadic tribes, long-houses for agrarian villages, and the two unique city sets; a tribe's **capital** carries a golden star overlay on the same art. (The art classes are hand-verified from the shipped sheets; the exact code site that selects the frame per tribe is not yet traced — TBD.)

ID	TRIBE	CLASS	GIFT GOOD	COLOR	SETTLEMENT
4	Incas	3 — Civilized (Cities)	Jewelled Relics	97	 Inca city
5	Aztecs	2 — Advanced (Cities)	Gold Bars	149	 Aztec city
6	Arawaks	1 — Agrarian (Villages)	Bone Jewelry	54	 village
7	Iroquois	1 — Agrarian (Villages)	Wood Carvings	87	 village
8	Cherokee	1 — Agrarian (Villages)	Turquoise	67	 village
9	Apache	0 — Semi-Nomadic (Camps)	Beads	111	 camp
10	Sioux	0 — Semi-Nomadic (Camps)	Beads	118	 camp
11	Tupi	0 — Semi-Nomadic (Camps)	Gems	71	 camp

Capital overlay:  *— the golden star drawn over the settlement art on each tribe's one capital.*

The id column is the village record's `settlement.owner` value — tribe index plus 4, a convention both the destroy path and the display path agree on. `@TRIBES` continues with eighteen reserve, name-only tribes that never instantiate (Maya, Toltecs, Kiowa, Hurons, Hopi, Navajo, Cheyenne, Cree, Algonquin, Powhatan, Delaware, Shawnee, Illinois, Chickasaw, Choctaw, Seminole, Mohicans, Zapotec). The class row "Capital" in NAMES `@LEVELS` is a settlement *type*, not a fifth tech level.

When a native speaks, the dialog portrait sheet is chosen by name — `IND<tribe><pose>` — via `pick_native_portrait()`: the speaker channel holds the tribe index (values above 7 switch to the King's portraits), and the pose digit is a banding of the tribe's current tension toward the viewing power (the banding helper's exact cut points are unresolved — TBD).

19.3 First contact

First contact, start to finish

native_events()

1. **Meet on land** — ships and wagons are refused — “We must contact the Indians on land first, Excellency.”
2. **The woodcut plays, once ever** — Inca → THE INCA NATION · Aztec → THE AZTEC EMPIRE · all others → MEETING THE NATIVES.
3. **The treaty offer** — “We generously offer you the land you now occupy… live with us in peace as brothers?” — Yes / No.
4. **Yes — peace for now** — anger starts from the map-creation seed: random(0..14) + 2·difficulty (human only), capped 20.
5. **No — war** — “Then the mighty {tribe} shall mercilessly drive you from our shores. Prepare for WAR!”

Greeting variants (worthy/ruthless tone) and a separate treaty text exist with untraced triggers TBD; no tribe-side “met” latch has been decoded TBD.

The sheet above is the whole sequence; the treaty text and the woodcut titles are quoted verbatim from the game script.

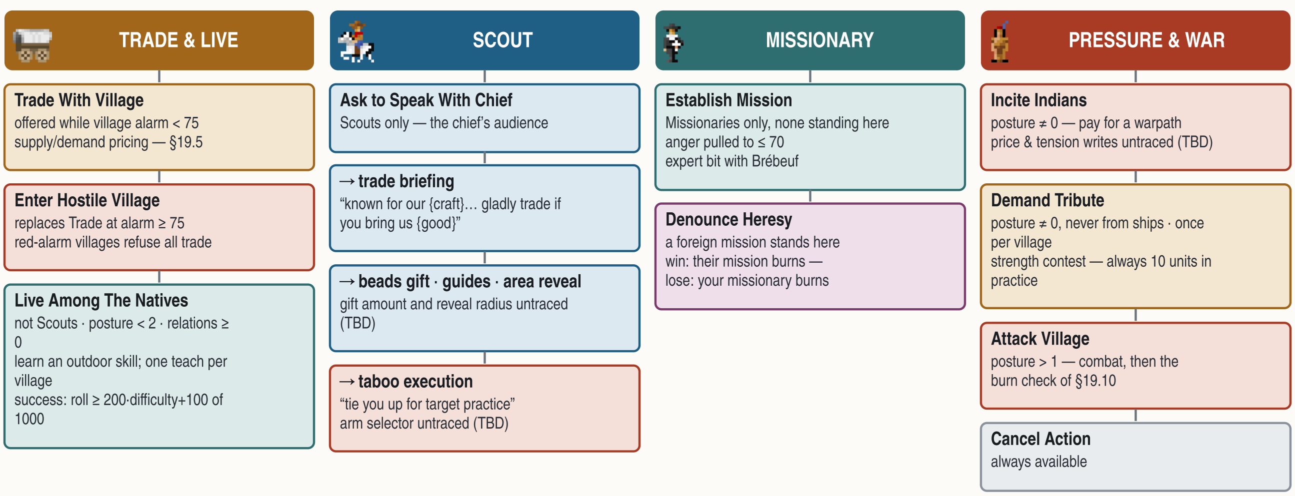
The tribe's starting disposition was already fixed at map creation, when `place_native_settlements()` rolled each tribe's per-power anger seed:

> **`starting anger(power) = random(0..14) + 2·difficulty`**
>
> `the +2·difficulty applies to HUMAN powers only; result saturates at 20`
>
> **EXAMPLE** at Conquistador (difficulty 2) a roll of 9 seeds the tribe's anger toward you at 9 + 4 = 13; toward an AI rival the same roll stays 9

Two greeting variants (a *worthy* tone and a *ruthless* tone) and a separate treaty text exist in the game's script with no traced trigger — the condition that selects them is TBD. There is also **no tribe-side "met" latch** in the decoded state (the European powers keep one in their relation matrix); how the engine remembers a tribe has been met is untraced — TBD.

19.4 Entering a village — the ten actions

The village menu — four routes through ten actions enter_village() · gates as decoded from the row-enable code



"Posture" is a traced-but-undecoded per-tribe state byte (TBD). Wagon trains that press an angry village vanish without a trace.

Moving a unit onto a village (first time: the *ENTERING INDIAN VILLAGE* woodcut, human players only) greets you with an attitude line — "*Your expedition has reached a %s of {tribe}…*" dressed as Happy / Medium / Savage / Bad / War — and opens the NAMES `@ACTIONS` menu. `enter_village()` is the only consumer of that menu; it enables rows like this:

#	ACTION	OFFERED WHEN
1	Trade With Village	village alarm toward you below 75
2	Enter Hostile Village	replaces Trade at alarm 75 and above
3	Establish Mission	unit is a Missionary and no mission stands here
4	Denounce Heresy of %F's Mission	a foreign power's mission stands here
5	Live Among The Natives	relations ≥ 0, tribe posture below 2, unit is not a Scout
6	Ask to Speak With Chief	unit is a Scout
7	Incite Indians	tribe posture nonzero
8	Demand Tribute	tribe posture nonzero, and the unit is not a ship

#	ACTION	OFFERED WHEN
9	Attack Village	tribe posture above 1
10	Cancel Action	always

("Tribe posture" is a per-tribe state byte the gate code reads; its storage is traced but its semantic — likely a peace/war stance — is not yet decoded, TBD.) Angry villages turn traders away outright: ships get "*mad at ships*", wagon trains get warned, then killed outright — "*Our wagon train has disappeared without a trace in {tribe} country.*"

The chief's audience (Scouts). Speaking with the chief runs through the same handler that later computes raze loot, `raze_settlement()`. Documented outcomes: the trade briefing ("*known for our {craft}. We would gladly trade if you bring us {good}…*" — built by sorting the village's live demand), the offer of **guides**, the **map-area reveal** ("tales of nearby lands"), the **beads gift** ("*Please take these valuable beads (worth {N})*"), a polite nothing, and — for taboo-breakers — "*You have broken sacred taboos… We shall tie you up for target practice.*" Which arm fires is steered by a sub-mode argument whose selector is untraced (TBD); the beads amount and the reveal radius are likewise TBD. The manual warns a scout can die in the attempt, "influenced by the mood of the tribe" — no odds are traced.

Living among the natives (training). `live_among_natives()` teaches only outdoor skills — Expert Farmer, Fisherman, Fur Trapper, Silver Miner, the three Master Planters, Seasoned Scout. Petty Criminals are refused ("even the greatest teacher must have suitable material"), colonists who already master a profession are refused, and **each village teaches exactly once** — the grant writes the profession into the unit and stamps the village's already-taught flag. Unskilled colonists learn at once on a difficulty roll:

```
learn succeeds when random(1..1000) ≥ 200·difficulty + 100
= 90 / 70 / 50 / 30 / 10 % from Discoverer down to Viceroy
```

EXAMPLE at Explorer (difficulty 1) a Free Colonist needs 300+ on the thousand-roll — 70 % — otherwise "you must live among us longer" and may retry

Which skill a given village offers is stored nowhere that has been mapped — the per-settlement skill field is TBD.

19.5 Trading with a village

A trade session

village_supply_demand() · haggle_trade()

1. **The village prices the moment** — demand & supply recomputed per good (§10 model); a capital doubles raw-goods demand and manufactured supply; "especially interested in…" = the sorted demand top.
2. **You sell** — offer = max(1, (seed·demand + 5·mood)·qty/100/2), seed = 2·(base − difficulty − want + mood + 4), mood = random(1..5).
3. **…or you buy** — ask = 200 (or (8−level)·50 for horses and manufactures) + market term + random(0..ask) − 4·surplus + 4·tension, floor 50.
4. **The haggle** — budget = random(1..rounds) + qty/4, rounds = min(3, (demand−want+4)/10) — counter-offers until the budget runs dry.
5. **Settle up** — gold moves 32-bit both ways · selling qty cools the village: alarm −qty, a 100-load zeroes it · muskets/horses sold arm the tribe (+1 lore at 25, +2 at 50) · tension −4.

Same visit: food-beg on a computed deficit, the corn gift on a fat surplus. The "let it be our gift" row's tension credit is untraced TBD; a village never buys the same good twice running (muskets excepted).

What the village wants is recomputed by `village_supply_demand()` — the full three-phase model (claimed-tile mask, 5×5 terrain scan, tribe-level supply and demand formulas) is documented in §10. Its outputs are two 16-good arrays, *demand* and *supply*, with two headline behaviours: a **capital doubles** its demand for raw goods (×1.5 for tools/muskets/trade goods) and doubles its supply of manufactures; and a village whose computed food demand outruns supply begs food from a visiting cargo ("*our people are hungry*"), while a fat surplus triggers the corn gift ("*accept this gift of corn*"). The chief's "*especially interested in…*" line names the top of the sorted demand list.

Selling. The village prices your cargo in one pass, then haggles:

```
sell offer = max( 1, ( max(0, seed·demand) + 5·mood ) · qty / 100 / 2 )
seed = 2·( base − difficulty − want + mood + 4 ), mood = random(1..5)
base is 6 for raw goods and 7 for manufactures, then per-good colour: Furs −random(0..7) ·
Muskets +(12 − tribe.muskets_known) · Horses +(10 − tribe.horses_known) · Trade Goods +1
want = the village's interest rank in the good, halved once its stock of it reaches 20,
forced to 0 for muskets and horses
```

EXAMPLE 100 Cloth to a village with demand-stock 10, mood 4, at Explorer: seed = 2·(7−1−2+4+4) = 24 → (24·10 + 20)·100/100 = 260 → halved = 130 gold offered

```
haggle budget = random(1..rounds) + qty/4
rounds = min( 3, (demand − want + 4)/10 )
```

EXAMPLE demand-stock 10, want 2 → rounds = 1; selling 100 units gives a budget of 1 + 25 = 26 — the village walks away when the budget is spent

Buying. The village prices its own goods the other way up:

```
buy price = max( 50, qty·ask/100 + ( difficulty + random(0..2) )·10 )
ask starts at 200 — for horses and manufactures it is (8 − tribe.level)·50 instead — then,
for silver and better: + market price · (2·difficulty + 15)
then + random(0..ask), − 4 · (village's surplus of the good), + 4 · (tribe's tension toward
you)
```

EXAMPLE 100 Horses from an Agrarian village (level 1): ask = 350, +34 market term, +200 rolled, −24 surplus, +52 tension = 612 → 612 + 20 = 632 gold

A closed deal moves the gold through your treasury (`power.gold`, 32-bit both ways, with an honest can't-afford check) and records the good in the village's trade memory. Selling has two quieter effects, both byte-traced: **every unit sold cools the village directly** — its alarm toward you drops by the quantity, and a full 100-load zeroes it outright — and **selling Muskets or Horses arms the tribe**: +1 lore at 25 units, +2 at 50 (horses also add a quarter of the load to the tribal herd). A −4 goodwill credit on the tribe's tension is also on record (its site sits inside the raid handler's range — flagged). The haggle script also carries a "*No, let the goods be our gift to you*" row; the gift's tension credit is untraced (TBD), though the original manual says gifts cool anger faster than sales. Two manual-attested behaviours round out the loop: a village will not buy the same good twice in a row (muskets excepted), and red-alarm villages refuse all trade.

19.6 Alarm and anger — everything that raises and lowers it

The anger system on one sheet

adjust_tension() · get_tension()

TRIBE TENSION — 0..100, clamped every update

Content → Uneasy → Restless → Angry → Hostile → War

0 calm 25 50 75 hostile 100 war

VILLAGE ALARM WORD — per village, per power

Calm → Trade→Hostile swap → Raids armed

0 calm 75 menu swaps to Enter Hostile 128 raids arm

HEATS

CAUSE	DELTA	NOTE
Drift, heating	`+1`	pressure counter crosses +8
Trespass, minor	`+1`	crossing tribal land
Trespass, moderate	`+2`	stamps village memory
Trespass, severe	`+3`	
Mission destroyed / expelled	`+comp`	amount TBD
War council joins the Crown	`+100`	post-Declaration; razing them guarantees it
Colony pressure nearby	`+frac`	fills the ±8 fractional meter
Burial-ground desecration	`+100`	the unit is executed
Ally's smite pact	`+100`	diplomacy row

COOLS

CAUSE	DELTA	NOTE
Drift, cooling	`−1`	quiet-turn credit
Normalization sweep	`−1`	decay loop over settlements
Successful trade	`−4`	goodwill after a closed deal
Mission founded	`−comp`	result capped at 70
War council, toward the King	`−100`	the same event's paired write
A standing mission, every turn	`−frac`	alarm −3×strength too
Pocahontas acquired	`reset`	all attitudes → Content

Both meters share one applier: positive deltas are halved for France and for a power that owns Pocahontas; results clamp 0..100. Scripted causes (roads, deforestation, missionaries, attacks, crowding) carry no traced delta TBD. Neighbour villages get clamped by a propagation tail whose semantics are undecoded TBD.

All tension changes flow through the single applier `adjust_tension()` — the sheet above is the complete traced ledger.

Beyond the tribe meter, each village's **alarm word** is what actually arms raids (threshold 128) and the Trade/Hostile menu swap (75). The applier's tail also **propagates alarm to neighbouring settlements of the same owner**, clamping their words to two fixed levels — the trigger and semantics of that clamp are still undecoded (TBD).

Six scripted anger notices (`@PISS0..5`) name the classic causes — road building, deforestation, foreign missionaries, unprovoked attacks, population pressure — but none carries a traced numeric delta (TBD). The original manual adds, at function level: colony size, building density and proximity raise alarm; muskets and cannon in a colony raise it; soldiers moved near a settlement raise it; a missionary working the settlement lowers it over time; everything is amplified when a capital is involved. The map shows the state as exclamation marks over the village, ramping pale green → blue → yellow → brown → red (a rival's alarm draws on that nation's colour).

At map generation each tribe also received the per-power **starting seed** of §19.3 — difficulty's only other finger on the scale is the attitude evaluator, which spots the human a harsher demand threshold than any AI.

19.7 Missions and conversion

The mission life cycle establish_mission() · attempt_conversion() · denounce_heresy()

1. **Found it** — mission byte = your power · village anger pulled to ≤ 70 · expert bit with Brébeuf (retroactive when he joins) · map cross in your colour.
2. **Roll each eligible turn** — convert fires when random(0..15) < tribe.level + 2, doubled expert — an Indian Convert appears at your colony; unplaced converts die of lost faith after 8 turns.
3. **Congress tuning** — Sepúlveda +4 conversion metric · las Casas −4, and on acquire every Convert → Free Colonist.
4. **How missions end** — war-footing tribes burn missions (anger +computed, formula TBD) — or a rival denounces: their converts burn your mission, or your missionary burns at the stake (roll TBD).

Establish Mission (Missionaries only, one mission per village) writes the founding power into the village's mission byte — with the **expert bit** set when the founding power holds **Jean de Brébeuf** in Congress; acquiring Brébeuf later retroactively upgrades every mission you own. Founding also applies the anger *reduction* of §19.6 (capped so the meter lands at 70 or below), and the map marks the village with a cross in the owner's colour (the manual: a brighter cross for expert missions).

Each eligible turn the mission rolls for a convert via `attempt_conversion()`:

```
convert fires when random(0..15) < tribe.level + 2
the threshold doubles with Jean de Brébeuf
```

EXAMPLE an Aztec village (level 2) converts on 4 of the 16 roll values — one convert roughly every 4 turns; 8 of 16 with Brébeuf seated

A success spawns an **Indian Convert** at your colony (the colonist class that also enjoys the +1 staple bonus in the fields, §11). Converts that fail to join a colony within **eight turns** are eliminated "for loss of faith". Two Founding Fathers tune the pipeline: **Juan de Sepúlveda** adds +4 to the conversion metric; **Bartolomé de las Casas** subtracts 4 — and, on acquisition, upgrades every existing Convert to a Free Colonist. The per-turn scheduler (exactly when "each eligible turn" fires) is untraced — TBD.

Losing a mission hurts twice: the tribe's computed anger *increase* of §19.6, and the church's outrage — "{*tribe*} *burn* {*power*} *missions!*" — when a war-footing tribe torches them. *Denounce Heresy* against a rival's mission resolves to one of two endings — your denouncing power wins ("*converts burn the* {*rival*} *mission and erect a new,* {*yours*} *one!*") or loses ("*Loyal worshipers burn the* {*missionary*} *at the stake!*"); the win/lose roll is untraced (TBD; the manual ties it to the two powers' standing with the tribe).

19.8 Demands, tribute, and hired wars

You demand tribute. `demand_tribute()` stages a strength contest — your regional military score against the tribe's, each side rolling `random(0..strength)` (Spain and Hernán Cortés owners count ×1.5) — and each village pays **once ever** (a one-shot flag). The demand is in **goods, not gold**, and the coded clamp collapses in practice:

```
tribute = clamp( raw, 10, min( 3·wealth-word + 10, 100 ) )
the wealth word is only ever written zero → the ceiling is always 10
```

EXAMPLE every successful tribute demand in the shipped game is exactly 10 units of the chosen good, taken from the village's ledger

Outcomes run from goods handed over, through "*our people are poor*", to the two refusals (laughter, or dark warnings).

They demand from you. War-footing tribes press claims: reparations in gold ("*The {tribe} people cry out for justice*" — trigger and amount untraced, TBD), the contents of passing **wagon trains** (hand them over or circle the wagons), or goods straight from a colony's stores (man the stockade / hand them over). Land, road-building and deforestation objections arrive with a **buy-off row** — pay the named compensation and the tribe withdraws the objection ("Very well, we withdraw our objection."). **Peter Minuit** in Congress zeroes all land payments.

Incite. A unit in the village (posture permitting) can pay the tribe to take the warpath against a rival — "*We will gladly drive the {rival} from our ancestral lands in exchange for {N}.*" The asking price's formula and the payment's tension writes are untraced (TBD — the +100/−100 pair once filed here belongs to the post-Declaration war council, §19.11); the manual names its three factors — your missions with the tribe, their attitude toward you, and their attitude toward the target.

Hired wars, the other direction. In a European diplomacy meeting you can pay a rival power to "*ruthlessly smite the heathen*":

```
smite fee = clamp( factor · their treasury / 2500, 10, 200 ) · 50
halved when the hired power carries a certain attribute bit (meaning untraced)
```

EXAMPLE hiring a power with factor 6 while they hold 30 000 gold: clamp(72, 10, 200) · 50 = 3 600 gold — and they declare war on the tribe

And an AI ally busy "subduing the notorious heathen {tribe}" may ask *you* to join — accepting instantly sets that tribe's tension toward you to war footing (+100).

19.9 Raids

Raid resolution

native_raid() · scan_raid_targets()

1 · The gate — a village's alarm word reaches 128

a roll of random(1..12) − 1 fires the raid when it reaches 3·K + 1

human bias the roll gains +(difficulty − 2) against a human colony · **K** untraced TBD

2 · Severity

random(1..4) — softened to a lower tier while turn < 40·(2 − difficulty)

0 the raiding party is wiped out on the approach · **early game** the first decades pull their punches

3 · Dispatch

1 stores looted · 2 havoc wreaked · 3 gold stolen · 4 a building or ship burned

stores the village banks the haul · **payloads** beyond the messages TBD · **vs a human colony** the INDIAN RAID woodcut plays

The five outcomes and their message families:

ROLL	OUTCOME	MESSAGE FAMILY
wiped out	the raiding party dies on the approach	"raiding party wiped out"

ROLL	OUTCOME	MESSAGE FAMILY
1	stores looted — goods stolen from the colony; the village banks the haul	@RAIDSTORES
2	havoc wreaked in the colony	@RAIDWREAK
3	gold stolen	@RAIDGOLD
4	a building burned — or a ship in port burned	@RAIDBURN / @RAIDSHIP

The concrete payloads behind wreak/gold/burn/ship beyond their messages are unmapped (TBD). A raid against a human colony plays the *INDIAN RAID* woodcut. When native war parties win in the field the aftermath messages take over — colonies overrun or burned outright, units lost with muskets or horses seized (feeding the tribe's armament state), or your soldiers driving them off. The early-game downgrade above is why the first decades stay survivable on the easier difficulties.

19.10 Attacking a village — burn, loot, and extinction

Attack resolution

raze_settlement() · remove_settlement()

1 · Battle

a roll between 1 and ATK + DEF — the attacker wins if it lands at or below ATK

DEF defence bonus, doubled for a capital · **village base defence formula** TBD · **manual** cities ≫ villages ≫ camps

2 · Attrition — each victorious attack

population − 1 — the village burns when its last point falls

so a size-8 village takes 8 lost battles to erase · **last point** a human attacker also stamps the tribe's avenge flag (§19.11) · **escape roll** random(0..100), 140 with a Seasoned Scout, big villages re-roll upward — the treasure branch (wiring TBD)

3 · Raze gold — paid the moment the village falls

three dice added together × a roll of random(1..6) × 4 × (size + 1)

each die random(1 .. 10 − difficulty) · **size** the village population (2026-05-30 ruling) · **capitals** exceed the ceiling — bonus TBD

DIFFICULTY	DISCOVERER	EXPLORER	CONQUISTADOR	GOVERNOR	VICEROY
Raze gold ceiling (size 21)	15,120	13,608	12,096	10,584	9,072

1. **March in** — posture must allow war — "Shall we attack the {tribe}, Your Excellency?"
2. **The battle** — random(1..ATK+DEF) ≤ ATK wins.
3. **Attrition** — each win takes one population; the last point burns the village.
4. **Raze gold** — straight to the treasury — no ×100, no Treasure unit on this path.
5. **Aftermath** — the last village falling wipes the tribe; a survivor's horse herd and lore scale n/(n+1).

Each victorious attack re-runs steps 2–3; a size-N village dies on its N-th lost battle.

Worked example — a size-8 Iroquois village, Explorer difficulty, turn 96

BATTLE	attack 6 vs defence 3, random(1..9) rolls 4	`4 ≤ 6, succeeds`
ATTRITION	the village's eighth lost battle — its last point falls	`pop 1 → razed`
RAZE GOLD	rolls 5+7+3 = 15, ×4, ×4, ×9	`2,160 gold`
AFTERMATH	Iroquois still hold 4 villages — no extinction	`herd × 4/5`
TENSION	unprovoked attack heats the tribe	`delta TBD`

How many attacks until it burns? The population *is* the counter (byte-located 2026-08-01, closing what an earlier scan wrongly declared absent). Inside the combat resolution, every battle the village loses does exactly one of two things:

- population above 1 → **population − 1**, with the village-shrinks message; or
- population at its last point → **the village is destroyed** — a human attacker also stamps the tribe's avenge flag (feeding the post-Declaration war council of §19.11) — and the removal path of this section runs.

So a size-8 village takes **eight lost battles** to erase. The separate roll inside `raze_settlement()` — `random(0..100)`, widened to 140 by a **Seasoned Scout**, biased upward for big villages (re-rolled while value/4 exceeds it), with a big-treasure branch at 75+ — governs the treasure/escape outcome of the raze; its exact wiring to the payout timing is the one piece still untraced (TBD).

The payout. When the village falls, the razing power collects gold on the spot:

The gold lands as a 32-bit credit in `power.gold` — **no ×100, and no Treasure unit on this path**. The *size* operand carries a documented conflict: the raw instruction reads the tribe's level byte, while the user-verified 2026-05-30 ruling identifies the true input as the village's population — the level read is the "Apache richer than Aztec" bug. Formula ceilings at size factor 21 run 15 120 / 13 608 / 12 096 / 10 584 / 9 072 from Discoverer to Viceroy — yet both captured **capital** razes exceed their ceiling, so a capital-only bonus exists whose magnitude is unmapped (TBD). The chief's execution scene ("*You have broken sacred taboos…*") is the scripted send-off. Treasure **units** — the wagons a Galleon hauls home — spawn only on the colony-combat path, with the King's cut on arrival governed by `cash_in_treasure()` (§18).

What the map does next. `remove_settlement()` retires the record: native units of the village are detached, the settlement table is compacted (every later record shifts down one slot, links repaired), the live count drops, and the tribe's settlement count falls. If it was the tribe's **last village**, the tribe's dead bit is set and "*The {tribe} tribe has been wiped out.*" announces the extinction; a surviving tribe instead has its **horse herd and horse lore** scaled down by n/(n+1) — the record scaled is the tribe's, not the dead village's (corrected 2026-08-01). (The debug menu's *Kill Indians* item drives exactly this machinery tribe-wide, greying out already-dead tribes.) Villages are born through `create_settlement()` — capped at 84 live records — and placed at map creation by `place_native_settlements()` from the dispersal chart.

19.11 The background economy — growth, herds, and war parties

One native turn — the invisible pass native_turn_driver() · tribe_turn() · settlement_tick()

1. **Before the powers move** — the native pass runs first, every turn, for each living tribe.
2. **War-council check** — post-Declaration only: anger ≥ 25 vs random(1..400), or the avenge flag — then 1 in 2·(5−difficulty)+1 → @INDIANGRUDGE: +100 vs you, −100 the Crown, missions burn, armory rescaled.
3. **Each village ticks** — growth counter += population; at 20 → grow toward the class cap, or field the replacement brave (armed by the armory, mounted for 50 horses).
4. **Missions cool, colonies heat** — alarm − 3×mission strength per turn; colony pressure pushes back; every ±8 of fractional anger becomes one visible tick.
5. **Stocks and herds** — trade balances drift to zero by class+1 per turn; the herd grows by horse lore, capped at 2·(tribe pop + 25).
6. **The braves move** — each war party takes up to 20 AI actions per turn; the brave decision logic is still undecoded (TBD).

Every turn — **before** the four European powers move — the engine runs the native pass: it walks each living tribe, then each of that tribe's villages, and does all of the following invisibly.

Villages grow toward a class cap. Each village accumulates its own population into a growth counter every turn; at 20 the counter resets and the village either grows by one (if below its cap) or replaces a fallen brave. The cap is set by the tribe's class:

CLASS	VILLAGES	THE CAPITAL
0 — Semi-Nomadic camps	3	4
1 — Agrarian villages	5	7
2 — Advanced cities (Aztecs)	7	10

CLASS	VILLAGES	THE CAPITAL
3 — Civilized cities (Incas)	9	13

growth fires every ceil(20 ÷ population) turns

big villages tick fast; a brave-replacement request preempts the growth

EXAMPLE a size-5 Iroquois village ticks every 4 turns and stops at 5 — its capital keeps going to 7

One brave per village. Map creation spawns exactly one Brave per village, linked to it. A village only builds another when **its own brave dies** — the unit-removal code stamps the rebuild-request flag, and the next 20-tick produces the replacement instead of growth. What marches out depends on the armory:

new brave: +armed if muskets lore > 0 (a musket is consumed 1 time in difficulty+1) · +mounted if the herd holds 50 (−50 horses)

EXAMPLE a Sioux village with musket lore 2 and herd 120 fields a Mounted Warrior — the herd drops to 70

Lore and herds. `tribe.muskets_known` is an armory that your musket sales (+1 at 25, +2 at 50), battlefield seizures, colony plunder and raids fill — and arming braves slowly drains. `tribe.horses_known` is a **breeding rate**: it is never spent, and the herd grows by it every turn, capped at 2·(tribe population + 25). Horse sales add a quarter of the load straight to the herd. This is why selling the tribes muskets and horses is remembered as the classic blunder — the goods become *war parties*.

Stocks fade. The per-good trade balance a village haggles against drifts back toward zero by (class + 1) per turn — a tribe forgets both gluts and shortages in a few dozen turns.

The war council (after the Declaration). Once independence is declared, each tribe checks every turn: if its anger toward the rebels is 25+ (rolled against random(1..400)) — or you ever razed one of its villages (the avenge flag) — then with odds 1 in 2·(5 − difficulty)+1 it holds the `@INDIANGRUDGE` war council: anger versus you jumps +100, anger toward the Crown drops −100, your missions in its villages burn, the armory is rescaled for war (musket lore ×4, the herd set to 25 per lore point) and the tribe is latched armed. This is the +100/−100 pair once mis-filed under "incite".

Missions and colonies push every turn. A standing mission cools its village continuously — alarm falls by 3× the mission strength each turn (strength: 1, or 4 for an expert mission, doubled in a capital; doubled again with las Casas seated, halved with Sepúlveda) — while nearby colony pressure heats it. Both feed a fractional per-power accumulator on the tribe; every ±8 in it becomes one visible ∓1 tension tick — that is the "drift" of §19.6. (The colony-pressure magnitude formula and the cool-down band helper are the remaining TBDs here.)

19.12 Aside — plundering European colonies

For contrast with raze gold: capturing a **European** colony (the colony-combat path, disabled during the War of Independence) transfers a population-weighted share of the victim's whole treasury:

plunder = captured colony's population × victim's whole treasury ÷ victim's total colony population

EXAMPLE taking a population-6 colony from a power holding 3 000 gold across 20 total population: 6 × 3000 ÷ 20 = 900 gold

Natives never collect plunder this way — a native victory over a colony runs the burn/overrun messages of §19.9 instead.